

jobs is the pattern of ingesting and processing large streams of data for near-real-time processing. Streams from services like Kafka and Kinesis can be processed by serverless functions, which can be scaled seamlessly to reduce latency and increase throughput of the system. This pattern can effectively handle spiky loads as well.

4. Automation/event-driven processing. Perhaps the first application of serverless computing was automation. Functions could be written to respond to certain alerts or events. These could also be periodically scheduled to augment capabilities for the cloud service provider through extensibility.

Kind of applications that are best suited for serverless architectures include mobile back-ends, data processing systems (real-time and batch) and web applications. In general, serverless architecture is suitable for any distributed system that reacts to events or process workloads dynamically based on demand. For example, serverless is suitable for processing events from Internet of Things (IoT) devices, large datasets (in Big Data) and intelligent systems that respond to queries (chatbots).

Serverless technologies

There are many proprietary and a few open source serverless technologies and platforms to choose from.

AWS Lambda is the earliest (announced in late 2014 and released in 2015) and the most popular serverless technology. Microsoft's Azure Functions supports a wider variety of languages and integrates with the company's Azure services. Google's Cloud Functions is currently in beta stage. One of the key open source players in serverless technologies is Apache OpenWhisk backed by IBM and Adobe. It is often tedious to develop applications directly on AWS, Azure, Google and OpenWhisk platforms. Serverless Framework is a popular solution that aims to ease application development on these platforms.

Many solutions (especially open source) focus on abstracting away the details of container technologies like Docker and Kubernetes. Hyper.sh provides a container hosting service wherein you can use Docker images directly in serverless style. Kubernetes from Bitnami, Fission from Platform9 and funktion from Fabric8 are serverless frameworks that provide an abstraction over Kubernetes.

Given that serverless is an emerging approach, technologies are still evolving. So you can see a lot of action in this space in years to come.

Challenges in going serverless

Despite the fact that a few large businesses are already powered entirely by serverless technologies, keep in

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